

ALEXANDER MATUS, Ph.D.

PHYSICAL APPLIED AI/ML SCIENTIST

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SUMMARY

Atmospheric Scientist with a PhD and over 15 years of experience engineering automated data pipelines, machine learning models, and image processing workflows for earth observation systems. Proven track record of taking legacy scientific algorithms and modernizing them into scalable, production-ready Python environments for NASA missions (CloudSat, CATS, CERES). Expert at integrating multi-sensor satellite imagery and multi-spectral datasets on High-Performance Computers (HPC) to perform atmospheric corrections, automate large-scale geospatial workflows, and support operational environmental monitoring.

TECHNICAL SKILLS

Scientific Software: Python (expert), R, Fortran, C++, Linux/Unix shell scripting (Bash), Git/Version Control
Image Processing: Multi-Sensor Data Fusion, Sensor Calibration, Image Analysis, Algorithm Optimization
Data Assimilation: Ensemble/Variational Methods, Quality Control (QC), Grid Mapping, Bias Correction
HPC/Computing: Linux Clusters, SLURM, Parallel Processing, Job Scheduling, Zarr, Dask, NetCDF
Earth Observation: Multi-spectral Imagery, Radar, Lidar, Radiometers, GEOS-5, ERA5, MERRA-2, CESM

EXPERIENCE

Assistant Research Scientist (40 hours per week)

10/01/2023 - Present

NASA Goddard Space Flight Center / UMBC

Greenbelt, MD

- **Multi-sensor Integration:** Combined massive satellite imagery and global model reanalysis data to study the global energy budget and close discrepancies in global precipitation models.
- **Technical Lead:** Lead studies for next-generation instrument requirements, successfully bridging the gap between scientific objectives and engineering constraints for future mission planning.
- **Operational Field Support:** Served as the onboard Instrument Scientist for major NASA airborne campaigns (IMPACTS, ACCLIP); managed real-time data and quality control on operational flights.

NASA Postdoctoral Fellow (40 hours per week)

10/01/2021 - 9/30/2023

NASA Goddard Space Flight Center

Greenbelt, MD

- **Algorithm Development:** Developed a new air quality estimation algorithm by combining satellite lidar data with neural networks (PyTorch/TensorFlow) and data assimilation, improving prediction accuracy.
- **Radiometric Bias Reduction:** Integrated satellite data into NASA's GEOS-5 model to significantly reduce spatial and temporal biases in air quality forecasting of surface-level particulate concentrations.
- **Project Leadership:** Served as PI for a \$170K research grant; led cross-disciplinary teams to integrate satellite-derived products into the GEOS-5 model for operational-style air quality monitoring.

CloudSat Algorithm Developer (40 hours per week)

8/01/2017 - 9/30/2021

University of Wisconsin-Madison/CIMSS

Madison, WI

- **Product Development:** Led the development of NASA's global CloudSat radiative flux products; modernized the production pipeline by migrating legacy compiled scripts into clean, scalable Python.
- **Workflow Automation:** Ran large-scale data processing and high-dimensional earth system simulations on Linux supercomputers using parallel processing and automated job schedulers.
- **Model Verification:** Conducted rigorous uncertainty analyses for global radiative effects to ensure product reliability, resulting in multiple peer-reviewed publications in high-impact journals.

EDUCATION

Ph.D., Atmospheric and Oceanic Sciences , University of Wisconsin-Madison	08/2017
M.S., Atmospheric and Oceanic Sciences , University of Wisconsin-Madison	08/2013
B.S., Meteorology (honors) , Pennsylvania State University	05/2009

* Transcripts available confirming 24+ semester hours in physical science/engineering

OPERATIONAL ROLES

Instrument Scientist | IMPACTS Airborne Campaign (10/2022 - 03/2023)

NASA Goddard Space Flight Center

Greenbelt, MD

- Managed real-time data streams and automated visualization tools on research aircraft for IMPACTS (Investigation of Microphysics and Precipitation for Atlantic Coast-Threatening Snowstorms) campaign.

Instrument Scientist | ACCLIP Airborne Campaign (07/2021 - 09/2022)

NASA Goddard Space Flight Center

Greenbelt, MD

- Directed real-time lidar calibration and radiometric response in support of ACCLIP (Asian Summer Monsoon Chemical and Climate Impact Project) aircraft field campaign at Osan Air Base, South Korea.

Algorithm Lead | CloudSat Algorithm Development (08/2016 – 09/2021)

University of Wisconsin-Madison/CIMSS

Madison, WI

- Directed version upgrades and operational implementation of CloudSat radiative flux products (2B-FLXHR-LIDAR) at the UW-Madison and Cooperative Institute for Meteorological Satellite Studies.

SELECTED PUBLICATIONS

Matus, A., R. Kramer, L. Oreopoulos, and N. Cho (2026). The Influence of the Atmospheric Radiative Budget on Global Precipitation. *Geophysical Research Letters*. (in review).

Matus, A., Nowottnick, E. P., Yorks, J., and da Silva, A. (2025). Enhancing surface PM_{2.5} air quality estimates in GEOS using CATS lidar data. *Earth and Space Science*, 12, e2024EA004078. doi: 10.1029/2024EA004078

Matus, A. and T. L'Ecuyer (2017). "The role of cloud phase in Earth's radiation budget." *Journal of Geophysical Research: Atmospheres*, 122, pp. 2559–2578, doi: 10.1002/2016JD025951

Matus, A., T. L'Ecuyer, et al. (2015). The role of clouds in modulating global aerosol direct radiative effects in spaceborne active observations and the Community Earth System Model. *Journal of Climate*. Vol. 28, pp. 2986–3003. doi: 10.1175/JCLI-D-14-00426.1

Total Citations: 700+ | **Publications:** 35

Awards: NASA Postdoctoral Fellowship, AOSS Best Poster, Lettau Award for Outstanding Thesis

AWARDS & HONORS

- Group Achievement Award IMPACTS Team**, NASA Goddard Space Flight Center (2025).
- Group Achievement Award ACCLIP Team**, NASA Goddard Space Flight Center (2024).
- NASA Postdoctoral Fellowship**, NASA Goddard Space Flight Center (2021).
- Lettau Award for Outstanding PhD Thesis**, UW-Madison (2017).
- AOSS Best Poster Award**, University of Wisconsin-Madison (2016).